Database Management System Lab — Lab Sheet 3

SQL Functions: Answers, MySQL Queries & Expected Outputs

DBMS used: MySQL / MySQL Workbench
Note: The lab sheet does not provide a fixed database schema or records. Therefore, the outputs below are based on a small sample database created specifically for this answer. If your college database has different records, the query syntax remains the same but the output values will change.

Sample Tables Used

Emp_ID	First_Name	Last_Name	Salary	Joining_Date
101	John	Doe	45000	2022-01-15
102	Alice	Smith	52000	2021-06-20
103	Rahul	Kumar	38000	2023-03-10
104	Priya	Sharma	61000	2020-11-05
105	David	Brown	47500	2022-08-18

Student_ID	Student_Name
1	Aarav
2	Priya
3	Rahul
4	Ananya
5	Karan

A. Number Functions

1. Demonstrate ABS() with a suitable example.

SQL Query:

```
SELECT ABS(-25) AS result;
```

Answer: ABS() returns the positive/absolute value of a number.

Expected Output:

result
25

2. Demonstrate ROUND() with a suitable example.

SQL Query:

```
SELECT ROUND(123.4567, 2) AS result;
```

Answer: ROUND() rounds a number to the specified number of decimal places.

Expected Output:

result
123.46

3. Demonstrate CEIL() and FLOOR().

SQL Query:

```
SELECT CEIL(12.34) AS ceil_value, FLOOR(12.34) AS floor_value;
```

Answer: CEIL() returns the smallest integer greater than or equal to the value; FLOOR() returns the largest integer less than or equal to it.

Expected Output:

ceil_value	floor_value
13	12

4. Demonstrate MOD() with a suitable example.

SQL Query:

```
SELECT MOD(17, 5) AS remainder;
```

Answer: MOD() returns the remainder after division.

Expected Output:

remainder
2

5. Calculate the rounded salary of employees.

SQL Query:

```
SELECT Emp_ID, First_Name, ROUND(Salary, -3) AS Rounded_Salary FROM Employees;
```

Answer: The salary is rounded to the nearest thousand using ROUND(Salary, -3).

Expected Output:

Emp_ID	First_Name	Rounded_Salary
101	John	45000
102	Alice	52000
103	Rahul	38000
104	Priya	61000
105	David	48000

B. Aggregate Functions

6. Find the total number of students using COUNT().

SQL Query:

```
SELECT COUNT(*) AS Total_Students FROM Students;
```

Answer: COUNT(*) counts all rows in the Students table.

Expected Output:

Total_Students
5

7. Find the total salary of all employees using SUM().

SQL Query:

```
SELECT SUM(Salary) AS Total_Salary FROM Employees;
```

Answer: SUM() adds all salary values.

Expected Output:

Total_Salary
243500

8. Find the average salary using AVG().

SQL Query:

```
SELECT AVG(Salary) AS Average_Salary FROM Employees;
```

Answer: AVG() returns the arithmetic mean of the salary column.

Expected Output:

Average_Salary
48700.0000

9. Find the highest salary using MAX().

SQL Query:

```
SELECT MAX(Salary) AS Highest_Salary FROM Employees;
```

Answer: MAX() returns the largest salary value.

Expected Output:

Highest_Salary
61000

10. Find the lowest salary using MIN().

SQL Query:

```
SELECT MIN(Salary) AS Lowest_Salary FROM Employees;
```

Answer: MIN() returns the smallest salary value.

Expected Output:

Lowest_Salary
38000

C. Character Functions

11. Demonstrate UPPER() on student names.

SQL Query:

```
SELECT Student_Name, UPPER(Student_Name) AS Upper_Name FROM Students;
```

Answer: UPPER() converts alphabetic characters to uppercase.

Expected Output:

Student_Name	Upper_Name
Aarav	AARAV
Priya	PRIYA
Rahul	RAHUL
Ananya	ANANYA
Karan	KARAN

12. Demonstrate LOWER() on student names.

SQL Query:

```
SELECT Student_Name, LOWER(Student_Name) AS Lower_Name FROM Students;
```

Answer: LOWER() converts alphabetic characters to lowercase.

Expected Output:

Student_Name	Lower_Name
Aarav	aarav
Priya	priya
Rahul	rahul
Ananya	ananya
Karan	karan

13. Demonstrate LENGTH() on student names.

SQL Query:

```
SELECT Student_Name, LENGTH(Student_Name) AS Name_Length FROM Students;
```

Answer: LENGTH() returns the number of bytes in a string in MySQL; for these ASCII names it equals the character count.

Expected Output:

Student_Name	Name_Length
Aarav	5
Priya	5
Rahul	5
Ananya	6
Karan	5

14. Demonstrate CONCAT() using first and last names.

SQL Query:

```
SELECT CONCAT(First_Name, ' ', Last_Name) AS Full_Name FROM Employees;
```

Answer: CONCAT() joins two or more strings into one string.

Expected Output:

Full_Name
John Doe
Alice Smith
Rahul Kumar
Priya Sharma
David Brown

15. Demonstrate SUBSTRING()/SUBSTR() with a suitable example.

SQL Query:

```
SELECT Student_Name, SUBSTRING(Student_Name, 1, 3) AS Short_Name FROM Students;
```

Answer: SUBSTRING() extracts a specified portion of a string. In MySQL, SUBSTR() is an alias commonly used for the same purpose.

Expected Output:

Student_Name	Short_Name
Aarav	Aar
Priya	Pri
Rahul	Rah
Ananya	Ana
Karan	Kar

D. Conversion Functions

16. Convert a number into a character/string value.

SQL Query:

```
SELECT CAST(12345 AS CHAR) AS String_Value;
```

Answer: CAST(... AS CHAR) converts the number to a character string.

Expected Output:

String_Value
12345

17. Convert a string into a numeric value.

SQL Query:

```
SELECT CAST('12345' AS UNSIGNED) AS Numeric_Value;
```

Answer: CAST(... AS UNSIGNED) converts a numeric string into an unsigned integer.

Expected Output:

Numeric_Value
12345

18. Convert a date value into a different format.

SQL Query:

```
SELECT DATE_FORMAT('2026-09-30', '%d-%m-%Y') AS Formatted_Date;
```

Answer: DATE_FORMAT() formats a date according to the specified format string.

Expected Output:

Formatted_Date
30-09-2026

19. Demonstrate CAST() with a suitable example.

SQL Query:

```
SELECT CAST('2026-09-30' AS DATE) AS Converted_Date;
```

Answer: CAST() converts a value from one data type to another supported type.

Expected Output:

Converted_Date
2026-09-30

20. Demonstrate CONVERT() where supported by the DBMS.

SQL Query:

```
SELECT CONVERT('2026-09-30', DATE) AS Converted_Date;
```

Answer: MySQL supports CONVERT(expr, type) for data-type conversion.

Expected Output:

Converted_Date
2026-09-30

E. Date Functions

21. Display the current date using a suitable date function.

SQL Query:

```
SELECT CURDATE() AS Current_Date;
```

Answer: CURDATE() returns the current date. The displayed value depends on the date on which the query is executed.

Expected Output:

Current_Date
2026-09-30

22. Extract the year, month and day from a date.

SQL Query:

```
SELECT YEAR('2026-09-30') AS Year, MONTH('2026-09-30') AS Month, DAY('2026-09-30') AS Day;
```

Answer: YEAR(), MONTH(), and DAY() extract individual date parts.

Expected Output:

Year	Month	Day
2026	9	30

23. Find the difference between two dates.

SQL Query:

```
SELECT DATEDIFF('2026-09-30', '2026-01-01') AS Days_Difference;
```

Answer: DATEDIFF() returns the number of days between two dates.

Expected Output:

Days_Difference
272

24. Add a specified number of days to a date.

SQL Query:

```
SELECT DATE_ADD('2026-09-30', INTERVAL 10 DAY) AS New_Date;
```

Answer: DATE_ADD() adds the specified interval to a date.

Expected Output:

New_Date
2026-10-10

25. Display employees' joining dates in a required date format.

SQL Query:

```
SELECT Emp_ID, First_Name, DATE_FORMAT(Joining_Date, '%d/%m/%Y') AS Joining_Date FROM Employees;
```

Answer: DATE_FORMAT() changes the displayed format without changing the stored date.

Expected Output:

Emp_ID	First_Name	Joining_Date
101	John	15/01/2022
102	Alice	20/06/2021
103	Rahul	10/03/2023
104	Priya	05/11/2020
105	David	18/08/2022

Important MySQL note: Functions can differ slightly between MySQL, Oracle, and SQL Server. This document is written specifically for MySQL/MySQL Workbench, as requested by the lab sheet. For Query 21, CURDATE() uses the actual current date when you run it, so your output may differ from the sample output shown here.